

MID-SEMESTER EXAMINATION, SEPTEMBER 2024

Course Code: ECECC08/EIECC08

Course Title: DIGITAL CIRCUITS AND SYSTEMS

Time: 1:30 Hours

Max. Marks: 15

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q.No.	Question	Marks	CO
1	A network has four inputs ABCD and four outputs WXYZ. ABCD represents a binary coded decimal digit. Further, WX represents the quotient and YZ represents the remainder when ABCD is divided by 4 (WX and YZ represent 2 bit binary numbers). Assume that the invalid inputs do not occur. (a) Draw the truth table. (b) Implement by using NAND gates only	1.5+1.5	CO1
2	The following Boolean function $f(w, x, y, z) = [(w + \bar{x})y + \bar{y}\bar{z}]$ (a) Draw the truth table (b) Implement it by using 8X1 MUX.	1.5+1.5	CO2
3	(a) Convert the number $(762)_8$ to 8421 code. (b) Perform subtraction using 2's complement $11011.001 - 1110.1101$	1.5+1.5	CO1
4	Simplify the following Boolean function F with the don't-care conditions d $F(A, B, C, D) = \Sigma (5, 6, 7, 12, 14, 15)$ and $d(A, B, C, D) = \Sigma (3, 9, 11)$ (a) Express in the SOP form (b) Express in the POS form	1.5+1.5	CO2
5	Given the function $f(A, B, C, D) = \Sigma_m (0, 4, 9, 10, 11, 12)$ and $f_1 = B \oplus D$, and $f = f_1 f_2 + f_3$. Determine the expressions of (a) f_2 (b) f_3	1.5+1.5	CO1